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Question 1

Q1.1 Four IT professional roles required for the project

Systems Analyst

A systems analyst acts as the bridge between the people who need a solution and the people who build it. In the ICT environment this role involves gathering requirements from stakeholders studying how users actually work and live, modelling processes and translating business or community needs into specifications that developers and engineers can act on.

In the Move With Me project the systems analysts role is visible in how the researchers worked to understand the mothers needs before committing to a design. The mothers explained that mobile data is rare and expensive with one saying she only had data "for 2-5 days a week" (Dewlok et al. 2025). The analysts job was to translate this constraint into a requirement: the cape had to work fully offline using Bluetooth Low Energy (BLE) with no dependence on internet connectivity or cloud services.

This role is essential for a community driven ICT project because rural communities are easily misunderstood by designers. A systems analyst ensures the communitys voice not the researchers assumptions becomes the blueprint, which exactly what kept the cape affordable, offline capable and locally repairable.

Software Developer

A software developer designs, writes, tests and maintains software. In the ICT environment this role covers everything from choosing the platform and programming language to writing clean code, debugging and releasing a working product that users can depend on.

In this project the developer was responsible for the PlayWithMe companion app, built on the Android Open Source Project (AOSP) platform in Kotlin a choice because Android holds an 84% market share in South Africa (Dewlok et al. 2025). The app gamifies movement through a cartoon character that demonstrates movements and reacts when the child copies them correctly communicating with the cape via an API over BLE.

This role is essential because hardware without software has little value to a child. The developer turned sensor data into a culturally relevant experience and had to keep the app free data free and compatible with phones the mothers already owned a key factor in whether the whole solution would actually be adopted.

Hardware / Embedded Systems Engineer

An embedded systems engineer. Builds the physical computing components that make devices work selecting microcontrollers, sensors, power sources and communication modules and writing the firmware that connects them.

In the Move With Me project this role is evident in the probes construction an Espressif ESP32 microcontroller managing data collection, an ADXL345 accelerometer detecting jumps and movement along three axes, Neopixel LED strips for feedback and an MP3 player with a speaker for audio cues (Dewlok et al. 2025). The engineer also implemented signal processing to filter out triggers calibrated movement thresholds for different childrens energy levels and ensured the electronics fit into a detachable breathable pocket so the cape could be hand washed.

This role is essential because in a low resource setting hardware failures cannot be fixed by ordering a replacement. The engineers choices low power BLE, a lithium ion battery locally available components directly determined whether the cape could survive real use in Sweet Waters and be repaired locally.

UX (User Experience) Designer / Researcher

A UX designer focuses on how real users experience a product, how intuitive it feels, whether it fits their context and culture and how the design process itself involves users. In ICT this role combines design thinking, prototyping and user research.

In this project the UX role appears in the use of a technology probe a functional prototype cape deliberately given to mothers to spark conversations about what the technology could be. It also appears in how the design was iterated based on user feedback mothers asked for more colourful lights the option to choose their own characters ("Maybe I want an animal or a robot instead of a person") animal sounds and even educational content like the cape reciting ABCs when the child jumps (Dewlok et al. 2025).

This role is essential in a community driven project because adoption lives or dies on whether people feel the product is theirs. The probe method removed the pressure of blank page ideation. Gave mothers genuine design agency, which is why the final capes were personalised, culturally grounded and owned by the community.

Q1.2 Three IT professional traits essential when working with rural mothers and quilters

Empathy and cultural sensitivity

Empathy is the ability to genuinely understand another persons circumstances, values and fears and in this project it was not optional. The researchers were working with mothers in an income rural community whose priorities (affordability, data costs, safety, cultural identity) were fundamentally different from what a Western design team might assume. This trait showed itself in small but telling moments, the quilters chose cultural fabrics for their grandchildrens capes rather than the Batman print and one mother noted her daughter "adores superman and batman" so she combined both (Dewlok et al. 2025). A team lacking empathy would have imposed a standard design. Instead the researchers honoured personalisation every cape came out different because they understood that a cape is not a device it is a gift from a mother to her child made with pride.

Communication

Co design only works if the researchers can explain concepts in a way that doesn't intimidate. Many of the mothers were initially afraid of the electronics one admitted, "In the beginning I was scared of connecting the wires to the battery. But now I am not scared it was really not hard" while another was honest enough to say "I don't like the batteries I am scared of them" (Dewlok et al. 2025). The teams willingness to give presentations repeat explanations and meet participants where they were turned anxious spectators into confident builders. Crucially communication ran both ways: the researchers also had to listen to suggestions like moving sound from the cape into the phone app, which cut costs dramatically. Without clear two way communication the mothers insights would never have reached the design.

Adaptability and humility

Perhaps the striking trait on display was the researchers willingness to change their own plans. They arrived with a prototype containing an MP3 player, a speaker and Neopixel LED strips and the mothers redesigned it in front of them proposing commercial LEDs that "bend easily so we can make much prettier capes " and suggesting a tiered product: "do you have little money right now lets build with the cheap lights do you have more money lets build with the fancy lights" (Dewlok et al. 2025). A rigid team would have defended its design. The Move With Me team adapted accepted that the community knew best and ended up with a cheaper more sustainable product because of it. In community driven ICT the professionals expertise is a tool to serve the communitys vision not the way around.

Q1.3 personal development plan

a) Two technical skills I must develop

The first is embedded systems development, specifically programming microcontrollers like the ESP32 using the Arduino IDE and C++. This is non-negotiable on a project of this kind: the entire cape runs on an ESP32 that reads sensor data from the ADXL345 accelerometer filters movements classifies jump types and triggers lights and sound in real time (Dewlok et al. 2025). During the December workshops the mothers themselves programmed microcontrollers and discussed the codebase as a graduate I cannot be the least capable person in the room when community members are learning to code.

The second is application development for constrained environments meaning Android development in Kotlin with an emphasis on offline first design, BLE communication and low resource consumption. The PlayWithMe app is the childs doorway to the experience, it receives movement triggers from the cape over BLE and displays the correct cartoon animation. It must run on the entry level Android phones that dominate South Africa without consuming data or battery (Dewlok et al. 2025). Building for constraint than for the latest flagship device is a specialised skill I would actively develop through this project.

b) Two professional development actions aligned with career planning

First I would join a body such as the ACM as a student or professional member and commit to its code of ethics. This aligns with my long term goal of working in ICT for development where conduct informed consent, respect for participants image rights honest reporting is directly tied to community trust. The Move With Me team obtained clearance gathered informed consent reaffirmed it repeatedly and even honoured the mothers request that their images not be blurred because they were proud of their work (Dewlok et al. 2025) that is the standard of practice I want to be held to.

Second I would pursue a mentorship and community engagement pathway, volunteering in co design workshops like the ones run in Sweet Waters and building a portfolio of ICT4D projects rather than only commercial ones. I would also aim to present or publish at conferences such as Communities & Technologies which would grow both my network and my credibility in the development research space.

c) How these skills and actions improve my employability

Development projects whether run by NGOs, government, universities or social enterprises need people who can wear hats: someone who can solder and debug firmware in the morning and facilitate a workshop with non technical mothers in the afternoon. Embedded systems and Kotlin/BLE skills are in demand generally but combined with demonstrated community engagement experience and professional body membership they signal to employers that I can deliver technology that works in the real world not just in a lab. The mothers cooperative idea producing and selling capes as a business shows how such projects create economic, graduates who understand this ecosystem become far more employable, for exactly the kind of ICT sector roles that South Africa needs most.

Question 2

The professional group: the Association for Computing Machinery (ACM)

The ACMs vision is to be the worlds leading computing society the organisation advancing computing as both a science and a profession. Its mission is to advance computing as a science and a profession: uniting educators, researchers and professionals to inspire dialogue share resources and address the fields challenges while promoting the standards and recognising technical excellence.

Its broad goals include advancing the science and art of computing through conferences and publications (like the CT proceedings in which this paper appears) supporting the growth of members through lifelong learning and career development; setting and promoting ethical and professional standards through the ACM Code of Ethics strengthening the professions collective voice in public policy and fostering diversity, equity and inclusion in computing worldwide.

The ACM promotes professionalism by giving researchers and practitioners a platform for peer reviewed publication, continuous learning and professional networking which collectively raise the quality and accountability of computing work. It promotes standards through its Code of Ethics and Professional Conduct which all members commit to and through initiatives such as its ethics education resources and its role in public policy debates on computings impact on society ensuring that technical excellence is matched by responsibility.

This principle holds that computing professionals should avoid consequences, especially where vulnerable people are involved in this case young children. Applied to co design, data gathering and prototype testing it meant the electronics had to be safe for children (separate breathable detachable pocket for components a design that made the cape hand washable) and that testing had to be closely supervised. It also shaped the decision to keep data local and offline no child or family data was pushed to the cloud reducing privacy and security risks in a context where households could not afford data.

The code requires professionals to protect information and only collect data with informed consent. In this project the team obtained clearance gathered written informed consent reaffirmed it regularly and explained exactly what data was being gathered and how it would be used. Notably when participants explicitly asked that their images not be blurred because they were proud of their work the researchers honoured that informed choice a textbook balance of confidentiality with autonomy.

Yes. I say that while acknowledging their limitations.The obvious value is credibility. When research appears in an ACM peer reviewed venue such as the CT proceedings readers know the work has been scrutinised by experts. For a study like Move With Me, which could easily be dismissed as a "community project," that stamp matters, it pushes community based research into the same serious conversation as mainstream systems research.

Second professional bodies set the ethical floor. The ACM Code of Ethics gave the research team a framework for consent, privacy and avoiding harm protections that matter enormously when your participants are mothers and their children. Without a body normalising these standards each researcher would set their own. The most vulnerable participants would bear the cost of the weakest ethics.

Third bodies provide infrastructure, conferences, journals, digital libraries, special interest groups and networks. The CT conference itself is where researchers studying community technology find each other. For career researchers especially that access to community and mentorship is often the difference, between persisting in research and quietly leaving it.

Fourth they advocate for the profession on access on research funding on policy questions like AI governance giving individual researchers a collective voice no one of them could achieve alone. The value is not evenly distributed. Membership and conference fees are expensive which quietly excludes researchers from resourced institutions. Ironically many of the people doing exactly the kind of rural Global South research the field claims to celebrate. Publication in prestige venues also tends to reward novelty over community impact and the peer-review process while valuable can be slow and opaque.

I believe the value outweighs these flaws and that the answer to them is engagement than abandonment. The ACMs role in legitimising the Move With Me research community voices, co design, rural constraints treated as class design problems is a good example of a professional body

amplifying work that genuinely needed amplifying. Bodies like the ACM are not perfect. For researchers they remain one of the most effective ways to turn careful ethical locally grounded work into knowledge the whole world can build on.

Question 3. Essay: IT, society and transformation

Technology that fits: what Move With Me reveals about ICT, society and transformation in South Africa.

South Africas rural communities sit on the side of almost every digital indicator: unreliable electricity, expensive and sporadic mobile data, low incomes and technology designed in and for wealthy urban markets. The Move With Me project is a lens through which to examine how ICT can either deepen these divides or help close them because it deliberately tackled the divide rather than ignoring it (Dewlok et al. 2025).

One clear positive impact of an ICT solution like the cape is on childrens health and early childhood development. Childhood inactivity and obesity are rising concerns in South Africa, where safe play spaces and well resourced Early Childhood Development centres are scarce and screen time is increasing. The cape converts movement into play. Flashing lights, sounds and a cartoon companion react when a child jumps. Making physical activity intrinsically rewarding than a chore. Because research shows that full body interaction improves not physical activity but also cognitive function and social skills (Dewlok et al. 2025) the impact reaches beyond fitness into school readiness.

A second positive impact is social empowerment. The project demonstrates how appropriate ICT can seed enterprise. The mothers began discussing a cooperative to produce and sell capes choosing price points. "Do you have little money right now let’s build with the cheap lights”. So that products could match real household budgets (Dewlok et al. 2025). When technology is locally repairable and locally made money circulates inside the community of leaking out to imported gadgets and the community gains skills as well as products.

The benefits of such projects are limited where technical skills are scarce and this is the uncomfortable part of the story. During the co creation sessions some mothers were afraid of the electronics; one never overcame her fear of batteries. This matters because technology that cannot be maintained locally degrades quickly. A broken sensor or flat battery with no one to diagnose it turns a R600 cape into waste. The mothers own spending on mobile data ranged from R50 to R350. An amount that leaves no room for repairs, replacements or technical support (Dewlok et al. 2025). Without upskilling communities remain dependent on experts, which recreates the dependency the project worked so hard to break. The smartphone requirement. The app needs a parents phone. Shows the gap. A household without a smartphone is excluded from the experience.

The project nevertheless supports transformation and inclusion in the ICT sector in a way few interventions do. Than treating rural people as end users of someone else’s product it treated them as designers and builders. Mothers learned to program microcontrollers discussed the codebase and made engineering trade offs (removing the speaker and MP3 module swapping Neopixels for strips) that cut costs while keeping functionality. The women moved from consumers of technology to producers of it. Exactly the shift South Africas ICT sector needs if it is to reflect the population it serves.

The decision to make the design open source extends this transformation allowing any community to adapt, repair and build on the work. The involvement of the mothers and quilters is precisely what made the cape suitable and freed it from design assumptions. A Western team would likely have shipped a cloud connected fitness wearable with a subscription app. Sensible and locally useless. Instead the community redirected the design at every step, BLE communication because "it is rare to have data because it is expensive" sound moved into the existing phone app because children were "so clever with phones" soft bendable wiring chosen over stiff strips because "I can make a flower or a heart" and traditional fabrics chosen by quilters alongside Batman prints

(Dewlok et al. 2025). The result was not a version of a Western product but a fundamentally different one. Local materials, local maintenance, local pride.

If IT professionals want to close the urban digital divide two practical initiatives stand out. The first is locally maintainable hardware and connectivity ecosystems. Professionals can. Support low cost devices assembled from locally stocked components (as this project did even confirming delivery to a shop within walking distance of the mothers' homes) coupled with community level connectivity solutions such as mesh networks or shared solar charging points. Each device that is cheap, repairable and offline converts one excluded household into a connected one and local assembly keeps the economic benefit in the community.

The second initiative is structured upskilling embedded in projects sustained hands on training like the six month co design readiness phase in which mothers received sensor kits, microcontrollers and laptops rather than once off computer literacy courses. When training is attached to an artefact the community owns skills stick confidence grows ("now I am not scared it was really not hard") and a local support base emerges that sustains the technology long after the researchers leave.

Together these initiatives address both sides of the divide. Access, to tools and the capability to use and maintain them. They convert ICT from something done to rural communities into something done by them. That ultimately is what transformation looks like.

Reference list

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